The Changing Anatomy of Vacancy Propagation: Industry Responses and Compositional Shifts

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Abstract

The timing of vacancy adjustment after a productivity innovation matters for labor-market propagation, but an aggregate vacancy response combines two objects: the behavior of vacancies within industries and the distribution of outstanding openings across those industries. Using the Conference Board's newspaper Help-Wanted Advertising Index for 1951Q1–2004Q2, we estimate productivity–vacancy–unemployment responses in progressively truncated samples. The impact response of vacancies generally becomes smaller as earlier decades are omitted, while positive point estimates at later horizons persist; the shortest-window estimates are imprecise. Using JOLTS data for 2001Q1–2019Q4, we then estimate manufacturing, private-services, and other-nonfarm responses to a common, recursively identified innovation in measured labor productivity. The manufacturing impact response exceeds the private-services response by 5.72 log percentage points (paired 90% pointwise bootstrap interval [3.93, 6.92]). Aggregating industry responses in vacancy levels and holding them fixed, we find that shifting five percentage points of the initial vacancy stock from private services to manufacturing raises the aggregate impact response by 0.29 log percentage points ([0.20, 0.35]), about 28% of the benchmark of 1.05. Our contribution is to link the historical change in response shape to an estimated sectoral contrast and an exact aggregation exercise. The exercise quantifies a channel through which industry composition may shape aggregate vacancy dynamics; identifying its historical contribution would require reconstructed historical vacancy-stock shares.

Keywords: Job vacancies; sectoral composition; labor-market propagation; manufacturing; measured productivity.

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1 Introduction

How much can differences in vacancy adjustment across industries, together with the composition of outstanding vacancies, alter the size and timing of the aggregate response to an innovation in measured labor productivity? The answer matters because aggregate vacancies help transmit productivity disturbances to unemployment and labor-market tightness. An aggregate vacancy response, however, combines two objects: the behavior of vacancies within industries and the distribution of outstanding openings across those industries. Fujita and Ramey (2007) study the delayed adjustment of vacancies and tightness following productivity innovations and show how vacancy-creation costs can generate propagation; related search-and-matching research emphasizes the role of vacancy creation in labor-market fluctuations (Shimer 2005; Elsby, Michaels, and Ratner 2015). A historical U.S. pattern brings the distinction between these two objects into focus: as earlier decades are omitted from a long newspaper-based vacancy series, the estimated impact response generally becomes smaller, while positive point estimates at later horizons persist. We document this pattern and treat it as motivation rather than as evidence of a compositional mechanism, then turn to an empirical comparison across modern industries and a conditional stock-aggregation exercise.

Our first finding is that the shape of the estimated aggregate vacancy response changes across historical U.S. estimation windows. Using the Conference Board's Help-Wanted Advertising Index, a newspaper-based vacancy proxy, for 1951Q1–2004Q2, we successively omit earlier decades while retaining a common endpoint and a common full-sample trend. Under our baseline two-lag specification, the estimated impact response is 2.52 log percentage points in the full window and 0.85 in the 1991-start window, while the corresponding point estimates three quarters after the innovation remain positive at 6.15 and 5.30. We interpret this pattern descriptively: the sequence of impact estimates is not monotonic, the shortest samples are imprecise, and marginal confidence intervals from overlapping windows do not directly test differences between windows. The comparison raises a question about the anatomy of the aggregate response, which we take up in the rest of the paper.

Our second finding is that, following the same recursively identified aggregate innovation, modern manufacturing vacancies respond much more strongly on impact than private-services vacancies. In pre-pandemic Job Openings and Labor Turnover Survey (JOLTS) data for 2001Q1–2019Q4, the estimated impact response is 6.19 log percentage points for manufacturing, compared with 0.47 for private services and 1.87 for other nonfarm industries. The paired manufacturing-minus-services impact gap is 5.72 log percentage points, with a

90% pointwise bootstrap interval of [3.93, 6.92]. The gap remains large over the first two quarters and narrows thereafter [**CHECK against Figure 2**], and positive early responses are not confined to manufacturing. These distinct sectoral paths offer a concrete reason why the industry mix of *outstanding vacancies*, not merely total employment or output, could shape the aggregate vacancy trajectory.

Our third finding is that, holding modern industry response schedules fixed, the aggregate response is sensitive, in economically meaningful amounts, to the initial composition of the vacancy stock. We aggregate the industry vacancy-stock responses in levels, normalized so that, by construction, they reproduce the directly estimated aggregate response under the reference initial weights. In that reference composition, manufacturing accounts for 6.52% of the nonfarm vacancy stock and private services for 79.61%. We then shift weight between manufacturing and private services while holding the initial other-nonfarm weight fixed. From a benchmark aggregate impact response of 1.05 log percentage points, a hypothetical five-percentage-point transfer from private services to manufacturing raises the response by 0.29 (paired 90% interval [0.20, 0.35]), about 28%, and a ten-point transfer raises it by 0.58 ([0.40, 0.71]), about 55%. Larger illustrative reallocations show more clearly how the full vacancy path and the *conditional* unemployment and tightness responses change. Because aggregation fixes the sign of the impact change given the estimated sectoral contrast, the substantive content lies in the magnitudes and the shape of the response paths.

Our analysis addresses a question about response dynamics that is distinct from the established debate over amplification in aggregate search models. Shimer (2005) documents a volatility problem for the baseline matching model, while Hagedorn and Manovskii (2008) and Hall and Milgrom (2008) examine, respectively, alternative calibrations and wage-setting mechanisms that change its quantitative behavior. Fujita and Ramey (2007), by contrast, focus on the speed of vacancy adjustment. We view our evidence as complementary to these structural accounts rather than as a test among them. We add an industry dimension to the propagation question: the relevant heterogeneity lies in vacancy-stock responses to a common aggregate innovation, and the relevant aggregation weights are shares of outstanding vacancies.

Existing industry-level evidence makes vacancy composition a natural object of study. Davis, Faberman, and Haltiwanger (2012) and Davis, Faberman, and Haltiwanger (2013) document substantial variation in job-filling rates across industries and in recruiting intensity across establishments, and Gavazza, Mongey, and Violante (2018) show in an equilibrium

model how recruiting effort transmits aggregate shocks to matching efficiency. Barnichon et al. (2012) decompose the post-recession shift in the Beveridge curve using industry-level JOLTS data, emphasizing widespread shortfalls in hires per vacancy. Şahin et al. (2014) investigate mismatch between vacancies and job seekers across labor markets, and Hobijn (2012) examines how the industry–occupation distribution of vacancies and hires bears on measured matching efficiency. These studies focus on recruiting and matching outcomes. We build on the broader idea that sectoral heterogeneity shapes aggregate dynamics by estimating manufacturing and services *vacancy-stock impulse responses to a common aggregate innovation* and recombining them under alternative initial vacancy weights. Our contribution is to link an empirical sectoral contrast directly to an explicit calculation of the aggregate vacancy response.

Two features of our setting shape how we interpret the historical comparisons: long-run vacancy measurement and possible changes in shock composition. On measurement, Barnichon (2010) explains why a newspaper-only help-wanted index loses coverage as recruiting moves online, and Elsby, Michaels, and Ratner (2015) survey the associated measurement and Beveridge-curve issues. Our historical vacancy proxy and the modern JOLTS *stocks of unfilled openings* differ in concept as well as in coverage. On shock composition, Abraham and Katz (1986) show why the comovement of sectoral dispersion and unemployment cannot, on its own, identify sectoral reallocation as the driving shock, and Garin, Pries, and Sims (2018) document shifts in the relative importance of aggregate and sectoral shocks for output, which may also contribute to changing aggregate dynamics. In addition, we identify an innovation in *measured labor productivity* rather than an externally identified technology shock. We therefore compare the shapes of the historical and modern responses rather than treating them as responses to the same disturbance measured on the same basis.

The composition exercise measures quantitative sensitivity rather than decomposing the historical pattern. Such a decomposition would require historical manufacturing shares of the vacancy stock, which have not been reconstructed and for which employment shares are not a substitute, as well as evidence that the modern sectoral response schedules held in earlier decades. In addition, the unemployment and tightness responses are *conditional* on the imposed vacancy path: we do not re-solve the vacancy equation under the altered unemployment trajectory. More broadly, our evidence points beyond the aggregate impact response to sectoral response timing and vacancy-stock composition. Reconstructing historical vacancy-stock shares would allow the historical pattern to be accounted for directly.

The paper proceeds from aggregate facts to sectoral responses and then to explicit aggregation. Section 2 describes the data, identification, exact aggregation of vacancy stocks, and the scope of inference. Section 3 presents the historical response shapes, Section 4 estimates the modern sectoral contrast, and Section 5 examines alternative initial vacancy compositions and the resulting conditional unemployment and tightness responses. Section 6 concludes.

2 Data and empirical design

2.1 Aggregate series, identification, and historical windows

We use quarterly U.S. data on measured labor productivity, vacancies, and unemployment, with two vacancy measures. The historical sample spans 1951Q1–2004Q2 and uses a quarterly series based on the Conference Board's Help-Wanted Advertising Index (HWI), a newspaper-based proxy for job vacancies (**conferenceBoardHWI**). The modern sample uses nonfarm JOLTS vacancy *stocks* for 2001Q1–2019Q4, ending before the pandemic; manufacturing and private services correspond to published JOLTS industry categories. JOLTS defines a job opening as a position that is unfilled on the last business day of the reference month and meets specific requirements: work is available, the job could start within 30 days, and the employer is actively recruiting from outside the establishment (U.S. Bureau of Labor Statistics n.d.). Openings therefore measure a stock of unfilled positions, distinct from the number of advertisements or the flow of newly created positions. Measured labor productivity is real GDP divided by CPS employment, and unemployment is the number of unemployed persons.[1]

The two vacancy measures serve different purposes. The HWI provides a long series for comparing the shape of the aggregate response across historical windows, whereas JOLTS provides the industry vacancy stocks needed for aggregation. The coverage of newspaper advertising also declines as recruiting moves online (Barnichon 2010). We therefore use the historical comparison as motivation and conduct the composition analysis entirely within the JOLTS sample.

[1] **CHECK: Real GDP is from the Bureau of Economic Analysis; CPS employment and unemployment are from the Bureau of Labor Statistics. JOLTS openings are [seasonally adjusted] monthly levels, converted to quarterly [averages/end-of-quarter values]; the monthly HWI is converted to quarterly [averages]. Private services comprise [list JOLTS supersectors].**

In both samples, we identify the productivity innovation recursively within a two-stage quarterly quasi-VAR, so it is an innovation in measured labor productivity rather than an externally identified technology shock. In the first stage, a VAR in measured productivity, aggregate vacancies, and aggregate unemployment, with productivity ordered first, identifies the productivity innovation and the productivity path it generates. This ordering allows vacancies and unemployment to respond to the innovation on impact, while their feedback to measured productivity operates only through lags. In the second stage, a vacancy–unemployment system conditions on its own lags, the first-stage productivity path, and the contemporaneous productivity innovation. We use two lags, selected once by the Bayesian information criterion (BIC) among lag orders one through four on a common effective sample from the modern period, and apply this baseline specification to the historical windows and to the modern industry system. Our baseline detrending applies a Hodrick–Prescott (HP) filter with smoothing parameter 100,000 to the log series, following Shimer (2005); we report linearly detrended estimates as a sensitivity check. [**WHERE ARE RESULTS FOR LINEARLY DETRENDED ESTIMATES? PUT AN APPENDIX AND REFER IT HERE.**] Impulse responses are normalized to an initial one-log-percentage-point increase in measured productivity and are reported in log percentage points (100 times the change in the natural logarithm), with horizon zero denoting impact. Because each historical window is estimated separately, the normalization fixes only the initial productivity movement, and subsequent productivity paths can differ across windows.

For the historical comparison, we hold the detrending fixed and successively remove earlier observations. For each detrending method, we detrend the full 1951Q1–2004Q2 series once and estimate windows beginning in 1951, 1961, 1971, 1981, and 1991, all ending in 2004Q2. Holding the full-sample trend fixed separates the effect of omitting observations from the effect of re-estimating the trend. Re-detrending each shorter sample defines a different exercise, which we report as a sensitivity check.[2]

[2]Our HWI series ends in 2004Q2, [**REASON, e.g., the last observation in the vintage available to us**], six quarters before the 2005Q4 endpoint in Fujita and Ramey (2007), and may differ from the data vintage they use. Our baseline specification also differs from theirs. Their VAR includes measured productivity, labor-market tightness, and the employment–population ratio and uses cubic detrending, whereas we estimate the productivity–vacancy–unemployment system described above. Like them, we use a vacancy–unemployment quasi-VAR to study propagation. Our estimates are therefore related to theirs but not directly comparable.

2.2 Modern industry responses and stock aggregation

The industry analysis combines an estimated contrast across industries with an exact aggregation identity that we impose. Let M, S, and O denote manufacturing, private services, and other nonfarm industries (construction, mining and logging, and government), whose vacancy stocks satisfy $V_t = V_{Mt} + V_{St} + V_{Ot}$. The reference weights $\bar{w}_i$ are the sample means of each industry's quarterly share of the nonfarm vacancy stock: 6.52%, 79.61%, and 13.87%, respectively. They are computed from vacancy counts rather than employment, and as averages of quarterly shares rather than ratios of pooled counts. We jointly estimate the responses of the relative log vacancy stocks, $100\log(V_{Mt}/V_{Ot})$ and $100\log(V_{St}/V_{Ot})$, to the common measured-productivity innovation. The relative-stock system uses the same second-stage structure as the aggregate system, conditioning on the first-stage productivity path, the contemporaneous innovation, and two lags of [**CHECK: the relative stocks and aggregate unemployment**]. The estimated relative responses determine the industry log responses $g_i(h)$ up to a common level at each horizon. We set that level so that aggregation in vacancy levels at the reference weights reproduces the directly estimated aggregate response $R_V(h)$:

$$R_V(h) = 100\log\left(\sum_{i\in\{M,S,O\}} \bar{w}_i \exp\left[\frac{g_i(h)}{100}\right]\right). \tag{1}$$

Because adding a constant to all three responses shifts the right-hand side one-for-one, equation (1) determines the common level uniquely and leaves the estimated contrasts, such as $g_M(h) - g_S(h)$, unchanged. The equation is therefore an accounting identity in vacancy levels, not an estimated result: the sectoral evidence comes from the relative responses, while the normalization makes the three industry paths consistent with the estimated nonfarm total. Aggregation in levels matters for two reasons. Because the log of a weighted sum differs from the weighted sum of logs, a share-weighted average of the industry log responses would only approximate $R_V(h)$. Level aggregation also lets the composition of the vacancy stock evolve along the response path, as industries with larger responses gain weight.

The composition experiment varies the initial weights while holding the estimated industry response schedules $g_i(h)$ fixed. For alternative initial weights $w = (w_M, w_S, w_O)$, the implied aggregate vacancy response is

$$R_V^{CF}(h;w) = 100 \log\left(\sum_{i\in\{M,S,O\}} w_i \exp\left[\frac{g_i(h)}{100}\right]\right), \tag{2}$$

where $w_O = \bar{w}_O$ and any change in the initial manufacturing share is offset one-for-one by private services, so that $w_S = \bar{w}_S - (w_M - \bar{w}_M)$. At the reference weights, equation (2) reproduces $R_V(h)$. As in equation (1), industry shares evolve along the response path. Figures 3 and 4 show the reference composition alongside illustrative initial manufacturing shares of 25%, 50%, and 75%, while Table 1 reports the effects of smaller transfers of five and ten percentage points. These weights are hypothetical and are not estimates of vacancy-stock shares in the historical sample.

The counterfactual unemployment and tightness responses are conditional on the reweighted vacancy path. We feed each counterfactual aggregate vacancy response $R_V^{CF}(h;w)$ into the fitted unemployment equation from the second-stage system, holding its estimated coefficients and the measured-productivity path fixed. Because vacancies enter the unemployment equation only with a lag, composition affects unemployment from horizon one onward. Let $R_U(h)$ denote the aggregate unemployment response and $R_U^{CF}(h;w)$ the counterfactual response this procedure produces. The implied tightness response is the difference between the log vacancy and unemployment responses, $R_\theta^{CF}(h;w) = R_V^{CF}(h;w) - R_U^{CF}(h;w)$, which reduces to $R_\theta(h) = R_V(h) - R_U(h)$ at the reference weights. Because the vacancy path is taken as given, we do not re-solve the vacancy equation under the altered unemployment trajectory. The resulting paths therefore trace the direct effect of vacancy composition on unemployment and tightness rather than a jointly determined vacancy–unemployment adjustment.

2.3 Inference

We report 68% and 90% pointwise intervals, which describe sampling uncertainty conditional on the specification. For the modern sample, the intervals come from 1,999 joint wild-bootstrap draws with HC2-adjusted residuals and Rademacher weights. Each draw re-estimates [**CHECK: both stages of**] the aggregate and industry systems [**CHECK: using common Rademacher weights across systems**]. Within each draw, we compute the manufacturing-minus-services contrast and the counterfactual-minus-benchmark differences $R_V^{CF}(h;w) - R_V(h)$, holding the reference weights $\bar{w}_i$ fixed, which yields the paired intervals reported in the text. The historical intervals also use 1,999 draws and condition on the full-

sample trend, so they do not include uncertainty about the trend itself; the linear-detrending results address the choice of detrending method. Because the intervals are pointwise, they characterize uncertainty at each horizon rather than over the full response path. The historical windows also overlap, so their intervals describe each window separately, and comparisons across windows are descriptive.

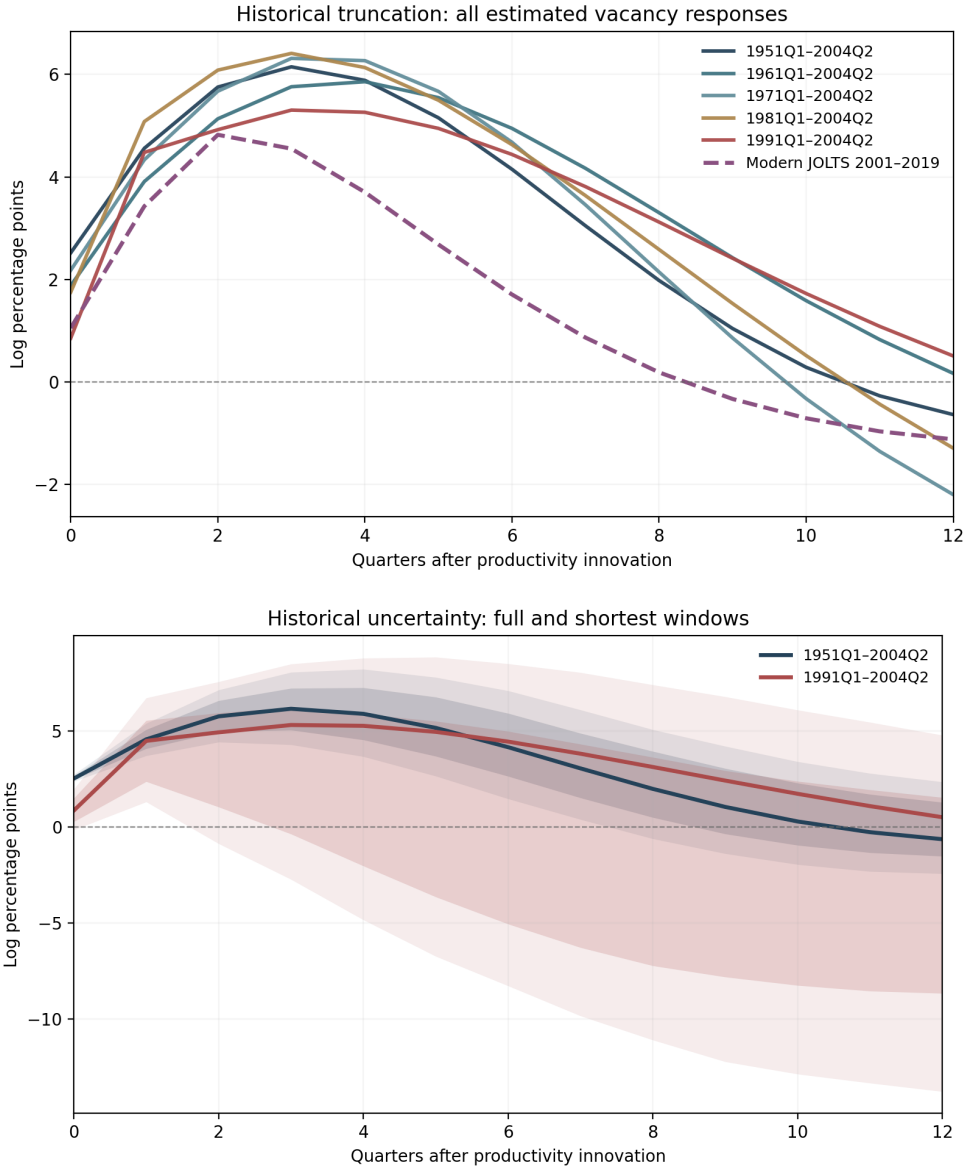
3 Historical aggregate response shapes

Shortening the historical estimation window generally reduces the estimated impact response of vacancies, while positive point estimates at later horizons persist. Figure 1 plots the aggregate vacancy responses for the five common-endpoint windows alongside the modern JOLTS benchmark. Under the baseline HP filter, the impact response is 2.52 log percentage points in the full sample and 0.85 in the 1991-start sample, while the corresponding responses three quarters after the innovation are 6.15 and 5.30. The intermediate windows show a similar pattern, although the impact estimates do not decline monotonically as the start date advances.

Sampling uncertainty grows considerably in the shorter windows. Three quarters after the innovation, the 90% interval is [4.28, 8.05] for the full sample and [−2.72, 8.48] for the 1991-start sample, so the short-window response is compatible with a wide range of shapes. The 1991-start window also spans the period in which online recruiting began to reduce the coverage of newspaper advertising (Barnichon 2010). [**CHECK: "The pattern is [similar] under linear detrending and when each window is re-detrended (Appendix ??).**] We therefore read Figure 1 as describing the shape of the aggregate response, a smaller impact response alongside positive responses at later horizons, rather than as evidence of a break at a particular date.

The modern JOLTS benchmark also rises after impact: vacancies increase by 1.05 log percentage points on impact and by 4.82 two quarters after the innovation [**CHECK**]. Because the vacancy measures, samples, and productivity paths differ, we use this benchmark as a reference for shape rather than as a direct comparison with the historical curves. Industry data allow us to examine the composition of the modern response directly. In particular, they distinguish a small response shared across industries from an aggregate response that averages markedly different industry paths, the question we take up in Section 4.

Figure 1. Historical aggregate vacancy responses



Notes: Upper panel: aggregate vacancy responses for the five common-endpoint windows, beginning in 1951, 1961, 1971, 1981, and 1991 and ending in 2004Q2, with the modern JOLTS benchmark (dashed). Lower panel: 68% and 90% pointwise intervals for the 1951-start and 1991-start windows, conditional on the full-sample trend. Responses are in log percentage points to a one-log-percentage-point innovation in measured productivity; horizon 0 denotes impact. Historical specification: two quarterly lags; HP smoothing parameter 100,000, applied once to the full 1951Q1–2004Q2 series. The JOLTS benchmark uses a different vacancy measure and a separately estimated productivity path.

4 Industry vacancy responses

Following the common measured-productivity innovation, manufacturing vacancies respond much more strongly on impact than private-services vacancies. Figure 2 shows impact responses of 6.19 log percentage points for manufacturing, 0.47 for private services, and 1.87 for other nonfarm industries. The paired manufacturing-minus-services gap is 5.72 log percentage points (90% interval [3.93, 6.92]), more than five times the aggregate impact response of 1.05. The gap is estimated directly from the relative-stock system, while the industry levels are normalized so that they aggregate to the estimated nonfarm response (Section 2.2).

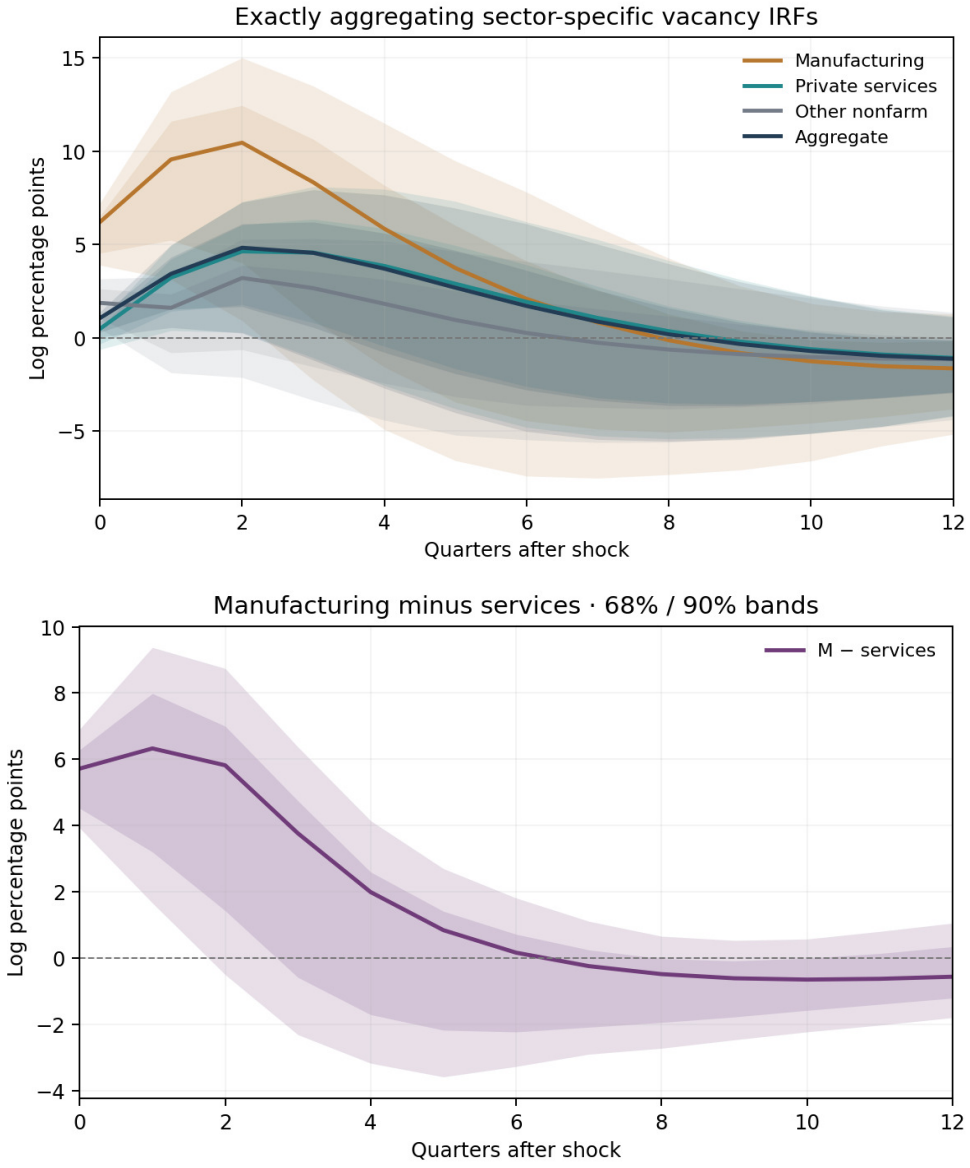
The industries differ in the magnitude and timing of adjustment rather than in whether they adjust at all. Manufacturing and private services both expand vacancies in the early quarters, so positive early responses are not confined to manufacturing. The gap remains large over the first two quarters and narrows thereafter, where it is also less precisely estimated [**MAYBE report the gap at horizon 3 and the first horizon at which its 90% interval includes zero?**].

Vacancy shares determine how these proportional responses enter the aggregate. Private services hold nearly four-fifths of the nonfarm vacancy stock, so their path carries a large weight even though their proportional impact response is small. Manufacturing's much larger response carries a weight of only 6.52%. On impact, manufacturing nonetheless contributes roughly 40% of the aggregate response. The modern aggregate response therefore averages markedly different industry paths rather than reflecting a small response shared across industries. Because the aggregate depends on these weights, the industry response schedules provide the empirical input for the composition exercise in Section 5.

5 Changing initial vacancy composition

Holding the industry response schedules fixed, a larger initial manufacturing share raises the aggregate impact response of vacancies and alters its subsequent path. Figure 3 applies equation (2) to compare the reference composition with illustrative initial manufacturing shares of 25%, 50%, and 75%, with each increase offset one-for-one by private services and the other-nonfarm share fixed at 13.87%. Under the reference composition, aggregate vacancies rise by 1.05 log percentage points on impact and by 4.82 two quarters after the

Figure 2. Industry vacancy responses and the manufacturing–services gap



Notes: Responses of log vacancy stocks, in log percentage points, to the common innovation in measured labor productivity, normalized to a one-log-percentage-point increase in productivity on impact; horizon 0 denotes impact. The upper panel shows the manufacturing, private-services, other-nonfarm, and aggregate responses. The lower panel shows the paired manufacturing-minus-private-services difference. Industry levels satisfy the aggregation identity in equation (1). Darker and lighter shading denote 68% and 90% pointwise intervals from 1,999 joint wild-bootstrap draws with HC2-adjusted residuals and Rademacher weights. Sample: 2001Q1–2019Q4; two quarterly lags; HP smoothing parameter 100,000.

innovation. With a 50% manufacturing share, these responses rise to 3.56 and 7.39, so the impact response more than triples. Figure 4 reports the paired differences between each counterfactual response and the reference response. The additional vacancy response increases with manufacturing's initial share. In the 50% scenario, it is 2.51 log percentage points on impact and 2.57 two quarters after the innovation, and it diminishes thereafter as the industry gap narrows [**CHECK: confirm against Figure 4; add the 90% intervals for these differences**].

Smaller transfers give the exercise a more interpretable scale. Table 1 shows that shifting five percentage points of the initial vacancy stock from private services to manufacturing raises the aggregate impact response by 0.29 log percentage points (paired 90% interval [0.20, 0.35]), about 28% of the reference response of 1.05. A ten-point transfer raises it by 0.58 ([0.40, 0.71]), about 55%. Two quarters after the innovation, the point estimates are similar in size, at 0.30 and 0.60, but they are small relative to the reference response of 4.82 and their intervals include zero, so the composition effect is most precisely estimated on impact. Because manufacturing's proportional impact response is larger, the aggregation formula fixes the sign of the impact effect; the empirical content lies in the size and precision of the industry gap and in the magnitude and shape of the reweighted aggregate paths.

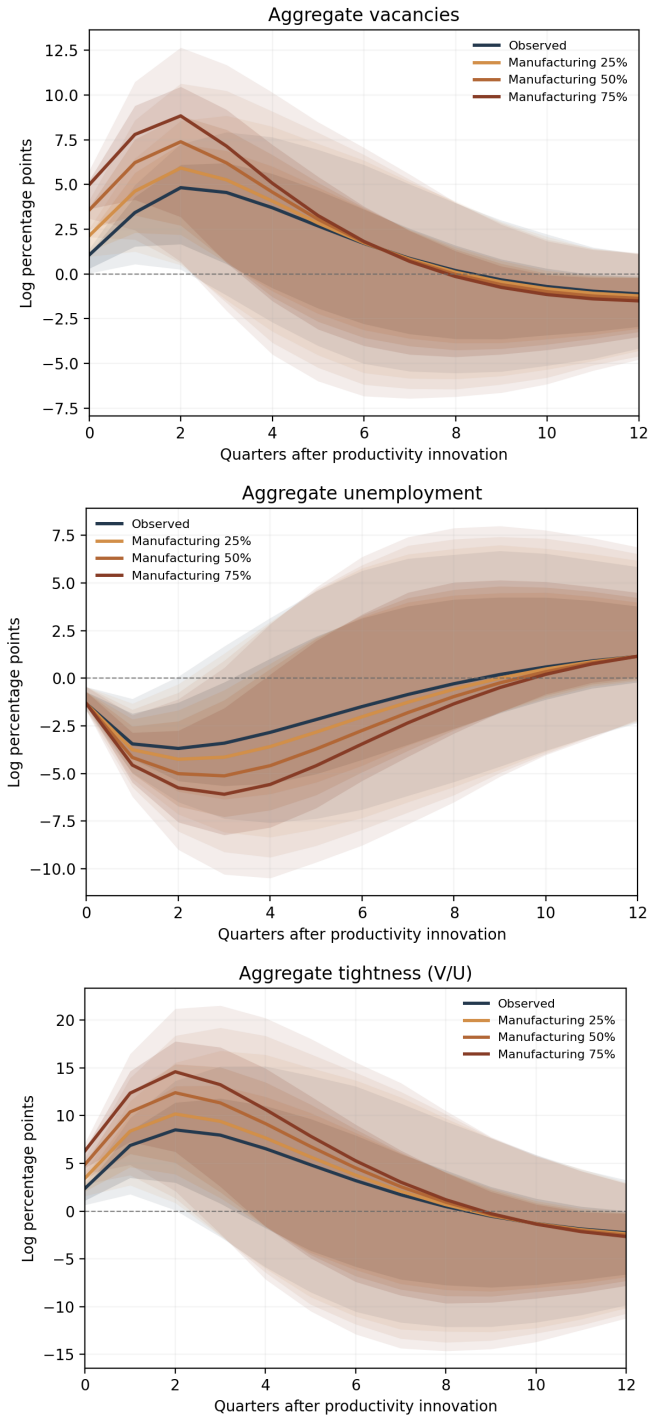
Table 1. Aggregate vacancy responses under smaller transfers from private services to manufacturing

Transfer to manufacturing (initial share)	Impact	Horizon 2
+5 pp (6.52% → 11.52%)	0.29 [0.20, 0.35]	0.30 [−0.02, 0.45]
+10 pp (6.52% → 16.52%)	0.58 [0.40, 0.71]	0.60 [−0.05, 0.90]

Notes: Differences between the counterfactual and reference-composition aggregate vacancy responses, in log percentage points, to a one-log-percentage-point innovation in measured productivity; horizon 0 denotes impact. Each transfer, in percentage points (pp) of the initial nonfarm vacancy stock, is offset one-for-one by private services; the other-nonfarm share is fixed at 13.87%. Industry response schedules and the productivity path are held fixed. Brackets contain paired 90% pointwise intervals from 1,999 joint wild-bootstrap draws with HC2-adjusted residuals and Rademacher weights, with differences computed within each draw. Sample and specification as in Figure 2.

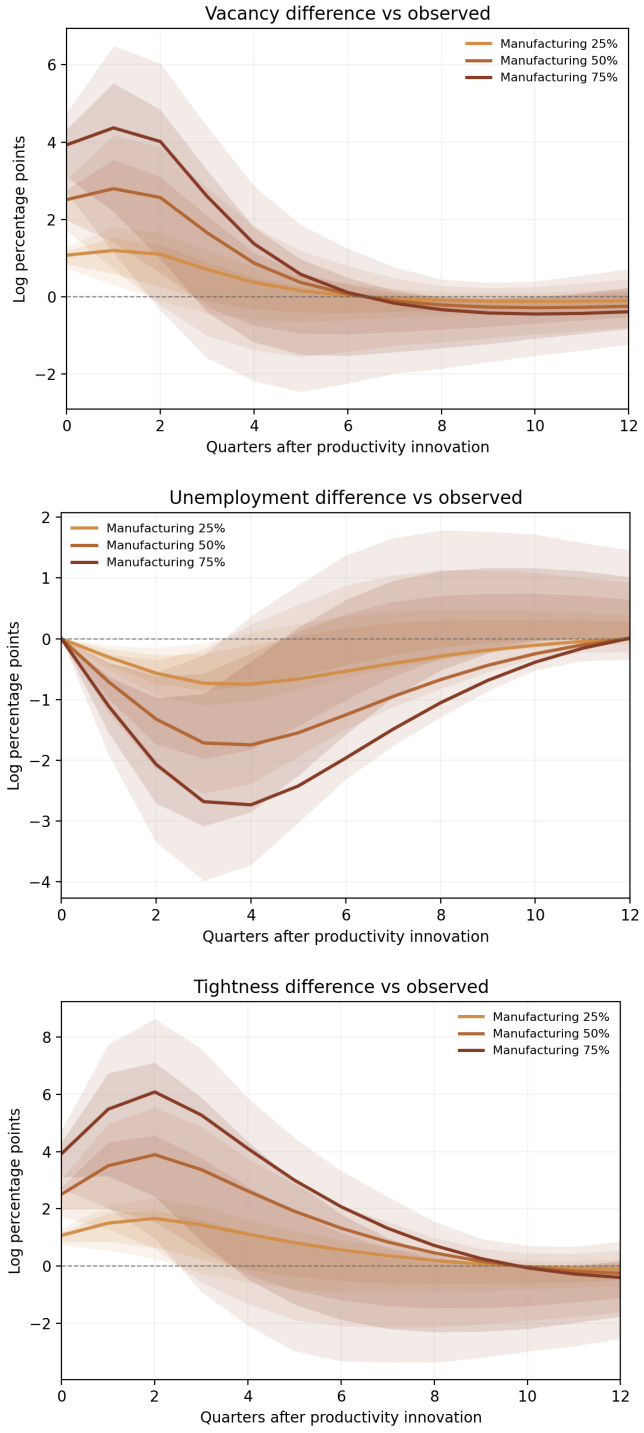
Through the fitted unemployment equation, the reweighted vacancy paths generate delayed, *conditional* movements in unemployment and tightness. Because vacancies enter the unemployment equation only with a lag, composition leaves the impact unemployment

Figure 3. Aggregate responses under alternative vacancy compositions



Notes: Responses of aggregate vacancies (top), conditional unemployment (middle), and conditional tightness (bottom), in log percentage points, under the reference composition and initial manufacturing shares of 25%, 50%, and 75%. Each increase in the manufacturing share is offset one-for-one by private services; the other-nonfarm share is fixed at 13.87%. Industry response schedules and the productivity path are held fixed. Unemployment is computed from the fitted unemployment equation conditional on each vacancy path, and tightness is the vacancy response minus the unemployment response (Section 2.2). Darker and lighter shading denote 68% and 90% pointwise intervals from 1,999 joint wild-bootstrap draws with HC2-adjusted residuals and Rademacher weights. Sample and specification as in Figure 2.

Figure 4. Response differences relative to the reference vacancy composition



Notes: Paired differences between each counterfactual response and the corresponding reference-composition response for aggregate vacancies (top), conditional unemployment (middle), and conditional tightness (bottom), in log percentage points. Initial manufacturing shares are 25%, 50%, and 75%, compared with a reference share of 6.52%; the other-nonfarm share is fixed at 13.87%. Industry response schedules and the productivity path are identical across scenarios. Differences are computed within each bootstrap draw. Darker and lighter shading denote 68% and 90% pointwise intervals from 1,999 joint wild-bootstrap draws with HC2-adjusted residuals and Rademacher weights. Sample and specification as in Figure 2.

response unchanged. In the 50% scenario of Figure 3, unemployment falls by 5.12 log percentage points three quarters after the innovation, compared with 3.41 under the reference composition. The paired difference of -1.72 (90% interval $[-2.55, -0.18]$) is about half the reference response. Tightness rises by 11.33 log percentage points in this scenario, compared with 7.96 under the reference composition, an increase of about 42% [**CHECK: add the 90% interval for the tightness difference**]. These paths trace how the reweighted vacancy responses pass through the fitted unemployment equation, holding vacancies fixed at their counterfactual paths (Section 2.2).

The size of the composition effect depends on two quantities: the estimated gap between industry responses and the share of the vacancy stock that is reallocated. Because the effect scales roughly in proportion to the transfer, even the smaller reallocations in Table 1 raise the aggregate impact response by more than a quarter. Vacancy composition can therefore change the size and early path of the aggregate response without any change in industry behavior.

6 Interpretation and conclusion

Industry differences in vacancy adjustment and the composition of outstanding vacancies can substantially change the aggregate response to an innovation in measured labor productivity. In the historical data, the estimated impact response of vacancies generally becomes smaller as earlier decades are omitted, while positive point estimates at later horizons persist; the shortest windows, however, are imprecise. In JOLTS data for 2001Q1–2019Q4, the manufacturing impact response exceeds the private-services response by 5.72 log percentage points, even though manufacturing holds only 6.52% of nonfarm vacancies and private services nearly four-fifths. Holding these industry response schedules fixed, shifting five percentage points of the initial vacancy stock from private services to manufacturing raises the aggregate impact response by 0.29 log percentage points, about 28% of the reference response, and a ten-point shift raises it by 0.58. Larger reallocations also change the subsequent vacancy path and, in the 50% manufacturing scenario, deepen the conditional decline in unemployment three quarters after the innovation by about half.

An aggregate vacancy response therefore combines industry behavior with the weights those industries receive in the vacancy stock. The industry contrast is the estimated input, and the composition exercise traces its aggregate implications conditional on fixed schedules.

Examining the full response path and its industry components reveals information that the aggregate impact response alone does not convey. Models of vacancy propagation should account for both the timing of industry responses and the vacancy shares through which those responses enter the aggregate.

How much industry reallocation contributed to the historical pattern remains an open empirical question. Answering it would require reconstructed historical vacancy-stock shares by industry, evidence on whether modern industry response schedules held in earlier decades, and a way to bridge the conceptual differences between newspaper help-wanted advertising and JOLTS openings. Our results establish vacancy composition as a quantitatively meaningful candidate source of differences in aggregate response shape, and these steps would allow its historical contribution to be assessed directly.

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